

TUGAS JURNAL MODUL 9



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1. Membuat project baru di visual studio dengan nama modul9_2311104065 menggunakan ASP.NET Core Web API. Lalu menambahkan file gitignore dan melakukan commit

```
C:\Windows\system32\cmd.exe
C:\Users\user\source\repos\KPL_Rengganis-Tantri-Pramudita_2311104065_S1SE-07-02\09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065> git add .
error: open("09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/FileContentIndex/b82959ff-0e10-4654-8ae9-caa6311be47a.vsix"): Permission denied
error: unable to index file '09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/FileContentIndex/b82959ff-0e10-4654-8ae9-caa6311be47a.vsix'
fatal: adding files failed

C:\Users\user\source\repos\KPL_Rengganis-Tantri-Pramudita_2311104065_S1SE-07-02\09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065> git add .

C:\Users\user\source\repos\KPL_Rengganis-Tantri-Pramudita_2311104065_S1SE-07-02\09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065>git commit -m "menambahkan file"
[main cfd8058] menambahkan file
21 files changed, 1638 insertions(+)
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/ProjectEvaluation/modul9_2311104065.metadata.v9.bin
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/ProjectEvaluation/modul9_2311104065.projects.v9.bin
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/ProjectEvaluation/modul9_2311104065.strings.v9.bin
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/DesignTimeBuild/.dtb.cache.v2
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/FileContentIndex/b82959ff-0e10-4654-8ae9-caa6311be47a.vsix
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/FileContentIndex/ef857add-e5ad-4bfb-8d30-0b5b011cf677.vsix
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/config/applicationhost.config
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/v17/.fud.cache.v2
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/v17/.suo
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/v17/DocumentLayout.backup.json
```

```
C:\Windows\system32\cmd.exe
ad-4bfb-8d30-0b5b011cf677.vsix
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/config/applicationhost.config
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/v17/.fud.cache.v2
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/v17/.suo
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/v17/DocumentLayout.backup.json
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/.vs/modul9_2311104065/v17/DocumentLayout.json
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/modul9_2311104065.sln
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/modul9_2311104065/.gitignore
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/modul9_2311104065/Controllers/WeatherForecastController.cs
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/modul9_2311104065/Program.cs
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/modul9_2311104065/Properties/LaunchSettings.json
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/modul9_2311104065/WeatherForecast.cs
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/modul9_2311104065/appsettings.Development.json
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/modul9_2311104065/appsettings.json
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/modul9_2311104065/modul9_2311104065.csproj
create mode 100644 09_API_Design_dan_Construction_Using_Swagger/modul9_2311104065/modul9_2311104065/modul9_2311104065.http

C:\Users\user\source\repos\KPL_Rengganis-Tantri-Pramudita_2311104065_S1SE-07-02\09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065>git push -u origin main
Enumerating objects: 37, done.
Counting objects: 100% (37/37), done.
Delta compression using up to 12 threads
Compressing objects: 100% (33/33), done.
Writing objects: 100% (35/35), 99.32 KiB | 1.68 MiB/s, done.
Total 35 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/rengganistan/KPL_Rengganis-Tantri-Pramudita_2311104065_S1SE-07-02.git
6425acc..cfd8058 main -> main
branch 'main' set up to track 'origin/main'.
```

2. Membuat folder Models yang berisi Movie.cs

```
namespace modul9_2311104065.Models
{
    public class Movie
    {
        public string Title { get; set; }
    }
}
```

```

    public string Director { get; set; }
    public List<string> Stars { get; set; }
    public string Description { get; set; }

    public Movie()
    {
        Stars = new List<string>();
    }
}

```

Penjelasan:

Kelas ini merepresentasikan data sebuah film dengan properti Title (judul), Director (sutradara), Stars (daftar pemeran), dan Description (deskripsi film). Properti Stars bertipe List<string>, yang berarti dapat menampung banyak nama pemeran. Konstruktor Movie() digunakan untuk menginisialisasi Stars sebagai list kosong agar tidak null saat objek dibuat. Kelas ini umum digunakan untuk menyimpan dan mengelola informasi film dalam aplikasi berbasis .NET.

3. Pada folder Controller isi dengan MovieController.cs

```

using Microsoft.AspNetCore.Mvc;
using modul9_2311104065.Models;

namespace modul9_2311104065.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    public class MoviesController : ControllerBase
    {
        private static List<Movie> movies = new List<Movie>
        {
            new Movie
            {
                Title = "The Shawshank Redemption",
                Director = "Frank Darabont",
                Stars = new List<string> { "Tim Robbins", "Morgan Freeman", "Bob Gunton", "William Sadler" },
                Description = "Two imprisoned men bond over a number of years, finding solace and eventual redemption through acts of common decency."
            },
            new Movie
            {

```

```

        Title = "The Godfather",
        Director = "Francis Ford Coppola",
        Stars = new List<string> { "Marlon Brando", "Al Pacino", "James Caan", "Diane Keaton" },
        Description = "The aging patriarch of an organized crime dynasty transfers control of his
clandestine empire to his reluctant son."
    },
    new Movie
    {
        Title = "The Dark Knight",
        Director = "Christopher Nolan",
        Stars = new List<string> { "Christian Bale", "Heath Ledger", "Aaron Eckhart", "Michael Caine"
    },
        Description = "When the menace known as the Joker wreaks havoc and chaos on Gotham,
Batman must accept one of the greatest psychological and physical tests of his ability to fight injustice."
    }
};

// GET /api/Movies
[HttpGet]
public ActionResult<List<Movie>> GetMovies()
{
    return movies;
}

// GET /api/Movies/{id}
[HttpGet("{id}")]
public ActionResult<Movie> GetMovieById(int id)
{
    if (id < 0 || id >= movies.Count)
    {
        return NotFound();
    }
    return movies[id];
}

// POST /api/Movies
[HttpPost]
public ActionResult<List<Movie>> PostMovie([FromBody] Movie movie)
{
    movies.Add(movie);
    return movies;
}

// DELETE /api/Movies/{id}
[HttpDelete("{id}")]
public ActionResult<List<Movie>> DeleteMovie(int id)
{
    if (id < 0 || id >= movies.Count)

```

```

    {
        return NotFound();
    }
    movies.RemoveAt(id);
    return movies;
}
}
}

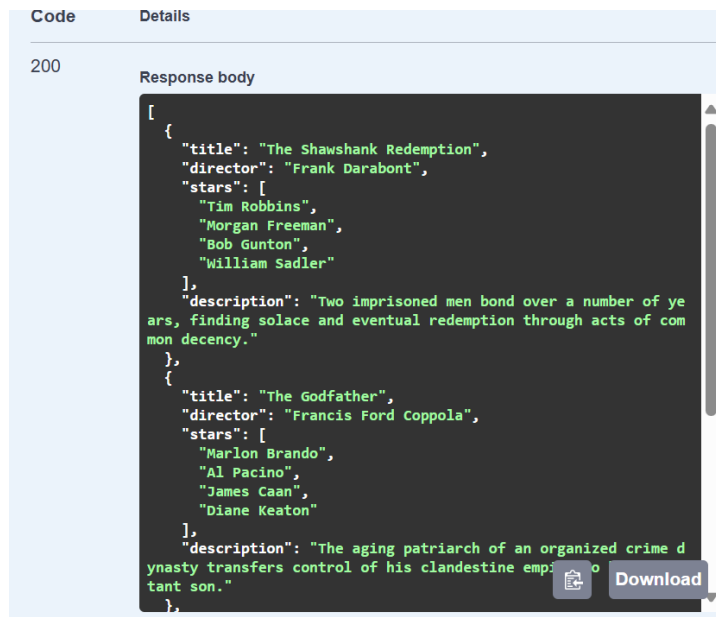
```

Penjelasan:

Kode di atas adalah implementasi **API controller** bernama **MoviesController** dalam ASP.NET Core, yang menyediakan endpoint untuk mengelola data film. Controller ini menggunakan routing `api/Movies` dan menyimpan data film secara lokal dalam list statis `movies`. Terdapat empat endpoint utama: GET untuk mengambil semua film, GET by id untuk mengambil film berdasarkan indeks, POST untuk menambahkan film baru, dan DELETE by id untuk menghapus film berdasarkan indeks. Setiap aksi menggunakan metode HTTP yang sesuai dan menangani validasi sederhana terhadap parameter id. Controller ini merupakan contoh RESTful API dasar untuk manajemen data film.

4. Mendemotrasikan web API

- a. Mencoba “GET /api/Movies” saat baru dijalankan yang mengeluarkan list film dari TOP 3 IMDB seperti pada tampilan berikut pada saat dicoba dengan menekan tombol “Try it out” dan tombol “Execute”



200

Response body

```
{
  "title": "The Godfather",
  "director": "Francis Ford Coppola",
  "stars": [
    "Marlon Brando",
    "Al Pacino",
    "James Caan",
    "Diane Keaton"
  ],
  "description": "The aging patriarch of an organized crime dynasty transfers control of his clandestine empire to his reluctant son."
},
{
  "title": "The Dark Knight",
  "director": "Christopher Nolan",
  "stars": [
    "Christian Bale",
    "Heath Ledger",
    "Aaron Eckhart",
    "Michael Caine"
  ],
  "description": "When the menace known as the Joker wreaks havoc and chaos on Gotham, Batman must accept one of the greatest psychological and physical tests of his ability to fight injustice."
}
]
```



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- b. Menambahkan Movie baru yaitu urutan ke-4 pada TOP IMDB list dengan memanggil API pada bagian “POST /api/Movies”

POST

/api/Movies

Parameters

Cancel

Reset

No parameters

Request body

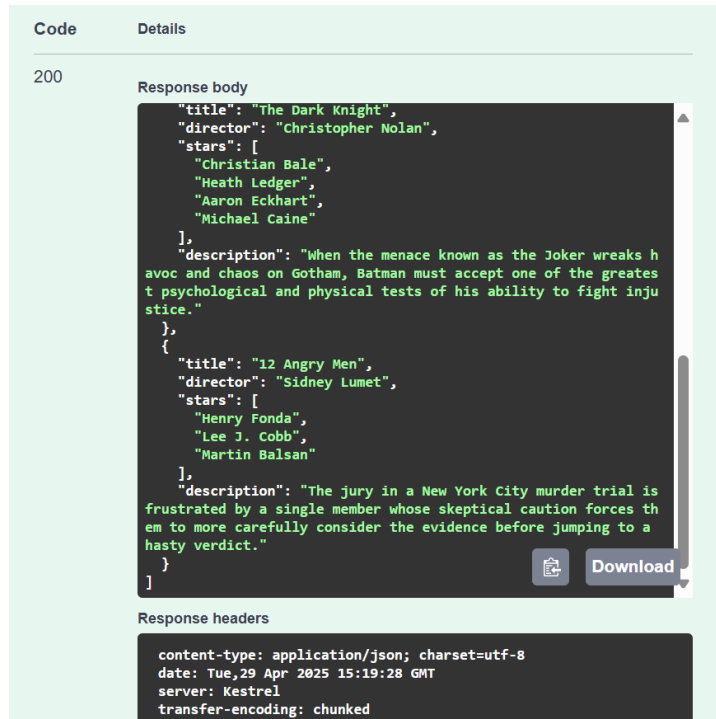
application/json

```
{
  "title": "12 Angry Men",
  "director": "Sidney Lumet",
  "stars": [
    "Henry Fonda",
    "Lee J. Cobb",
    "Martin Balsan"
  ],
  "description": "The jury in a New York City murder trial is frustrated by a single member whose skeptical caution forces them to more carefully consider the evidence before jumping to a hasty verdict."
}
```

Execute

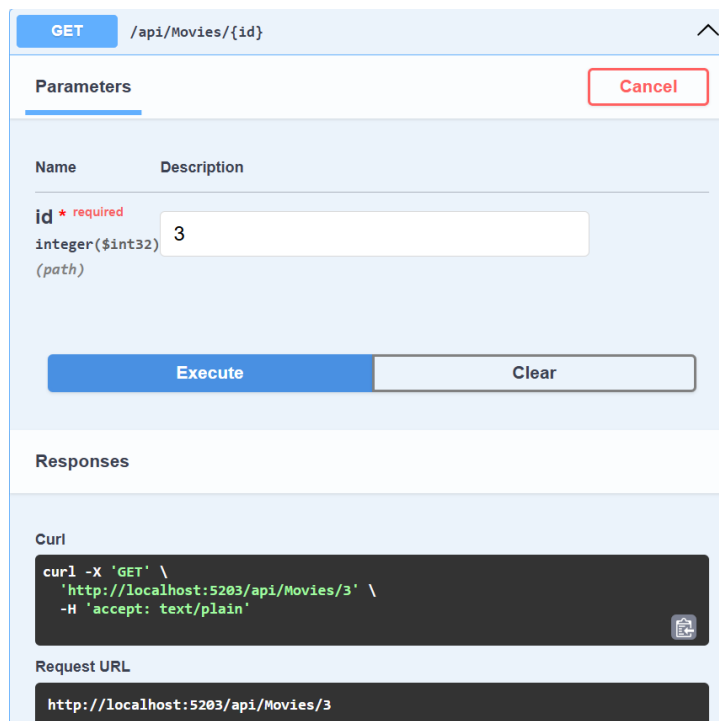
Clear

- c. Cek list/array dari semua Movie lagi dengan “GET /api/Movies”, pastikan Movie yang baru ditambahkan sebelumnya sudah ada:



```
200
Response body
{
  "title": "The Dark Knight",
  "director": "Christopher Nolan",
  "stars": [
    "Christian Bale",
    "Heath Ledger",
    "Aaron Eckhart",
    "Michael Caine"
  ],
  "description": "When the menace known as the Joker wreaks havoc and chaos on Gotham, Batman must accept one of the greatest psychological and physical tests of his ability to fight injustice."
},
{
  "title": "12 Angry Men",
  "director": "Sidney Lumet",
  "stars": [
    "Henry Fonda",
    "Lee J. Cobb",
    "Martin Balsan"
  ],
  "description": "The jury in a New York City murder trial is frustrated by a single member whose skeptical caution forces them to more carefully consider the evidence before jumping to a hasty verdict."
}
]
Response headers
content-type: application/json; charset=utf-8
date: Tue, 29 Apr 2025 15:19:28 GMT
server: Kestrel
transfer-encoding: chunked
```

- d. Mencoba meminta Movie dengan index 3, “GET /api/Movies/3” yang seharusnya mengembalikan Movie yang baru saja ditambah:



GET /api/Movies/{id}

Parameters

Name Description

id * required
integer(\$int32)
(path)

3

Execute Clear

Responses

Curl

```
curl -X 'GET' \
'http://localhost:5203/api/Movies/3' \
-H 'accept: text/plain'
```

Request URL

```
http://localhost:5203/api/Movies/3
```

Server response

Code	Details
200	Response body <pre>{ "title": "12 Angry Men", "director": "Sidney Lumet", "stars": ["Henry Fonda", "Lee J. Cobb", "Martin Balsan"], "description": "The jury in a New York City murder trial is frustrated by a single member whose skeptical caution forces them to more carefully consider the evidence before jumping to a hasty verdict." }</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Tue, 29 Apr 2025 15:21:45 GMT server: Kestrel transfer-encoding: chunked</pre>

Responses

Code	Description	Links
200	OK	No links

Media type

text/plain

e. Menghapus objek Movie dengan index ke-1 dengan “DELETE /api/Movies/1”

DELETE /api/Movies/{id}

Parameters

Cancel

Name	Description
id * required	
integer(\$int32)	1
(path)	

Execute Clear

f. Cek list/array dari semua Movie sekali lagi dengan “GET /api/Movies”, film dengan ranking kedua “Godfather” sudah tidak ada di list:

200

Response body

```
[
  {
    "title": "The Shawshank Redemption",
    "director": "Frank Darabont",
    "stars": [
      "Tim Robbins",
      "Morgan Freeman",
      "Bob Gunton",
      "William Sadler"
    ],
    "description": "Two imprisoned men bond over a number of years, finding solace and eventual redemption through acts of common decency."
  },
  {
    "title": "The Dark Knight",
    "director": "Christopher Nolan",
    "stars": [
      "Christian Bale",
      "Heath Ledger",
      "Aaron Eckhart",
      "Michael Caine"
    ],
    "description": "When the menace known as the Joker wreaks havoc and chaos on Gotham, Batman must accept one of the greatest psychological and physical tests of his ability to fight injustice."
  }
]
```

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200

Response body

```
{
  "title": "The Dark Knight",
  "director": "Christopher Nolan",
  "stars": [
    "Christian Bale",
    "Heath Ledger",
    "Aaron Eckhart",
    "Michael Caine"
  ],
  "description": "When the menace known as the Joker wreaks havoc and chaos on Gotham, Batman must accept one of the greatest psychological and physical tests of his ability to fight injustice."
},
{
  "title": "12 Angry Men",
  "director": "Sidney Lumet",
  "stars": [
    "Henry Fonda",
    "Lee J. Cobb",
    "Martin Balsan"
  ],
  "description": "The jury in a New York City murder trial is frustrated by a single member whose skeptical caution forces them to more carefully consider the evidence before jumping to a hasty verdict."
}
]
```

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5. Melakukan commit terakhir

```
C:\Users\User\source\repos\WPL_Rengganis-Tantri-Pramudita_2311104065_51SE-07-02\09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065>git add .

C:\Users\User\source\repos\WPL_Rengganis-Tantri-Pramudita_2311104065_51SE-07-02\09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065>git commit -m "menambahkan API Movies"
[master f34285b] menambahkan API Movies
18 files changed, 236 insertions(+), 59 deletions(-)
create mode 100644 09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065\.\vs\modul9_2311104065\FileContentIndex\85a83b68-3b69-4504-aad8-bd6d38559091.vsix
create mode 100644 09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065\.\vs\modul9_2311104065\FileContentIndex\710b31cb-f37b-4aed-b498-c37c62dd2eb0.vsix
create mode 100644 09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065\.\vs\modul9_2311104065\FileContentIndex\9108291c-2ad2-4e51-86f6-a742a73c57e3.vsix
create mode 100644 09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065\.\vs\modul9_2311104065\FileContentIndex\b12373bf-bbcb-4522-8e9e-476c88ef1d39.vsix
delete mode 100644 09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065\.\vs\modul9_2311104065\FileContentIndex\b82959ff-0e18-4650-ba59-caa311be07a.vsix
create mode 100644 09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065\modul9_2311104065\Controllers\MovieController.cs
delete mode 100644 09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065\modul9_2311104065\Controllers\WeatherForecastController.cs
create mode 100644 09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065\modul9_2311104065\Models\Movie.cs
delete mode 100644 09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065\modul9_2311104065\WeatherForecast.cs

C:\Users\User\source\repos\WPL_Rengganis-Tantri-Pramudita_2311104065_51SE-07-02\09_API_Design_dan_Construction_Using_Swagger\modul9_2311104065>git push -u origin main
Enumerating objects: 46, done.
Counting objects: 100% (46/46), done.
Delta compression using up to 12 threads
Compressing objects: 100% (25/25), done.
Writing objects: 100% (23/23), 40.18 KiB | 1.25 MiB/s, done.
Total 28 (delta 8), reused 1 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (8/8), completed with 7 local objects.
To https://github.com/rengganis-tantri-pramudita/WPL_Rengganis-Tantri-Pramudita_2311104065_51SE-07-02.git
```